Tate's Thesis

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Contents

1 Local Functional Equation

To treat the local field of number fields, we fix the notation for sake of convenience.

Definition 1.1. Let F be a field, then we call F is a local field if it is one of the following types:

1. Archimedean local field: $\mathbb{R}$ or $\mathbb{C}$;
2. Non-archimedean local field: a finite extension of $\mathbb{Q}_p$

Equivalently, a local field is a completion of a number field at a place, and it is non-archimedean if the place is finite.

In the case of non-archimedean local field, we use the following notation:

$$\begin{aligned} \mathcal{O} &: \text{ the valuation ring} \\ \mathcal{O}^\times &: \text{ the unit group of } \mathcal{O} \\ \mathfrak{p} &: \text{ the maximal ideal of } \mathcal{O} \\ \varpi &: \text{ a uniformizer} \\ \mathfrak{d} &: \text{ the different of } F/\mathbb{Q}_p \end{aligned}$$

Definition 1.2. Let F be a local field, then $f : F \to \mathbb{C}$ is called a Schwartz-Bruhat function if it is one of the following cases:

- When F is archimedean, f is a Schwartz function, which is a smooth and rapidly decreasing function of class C^∞.
- When F is non-archimedean, f is locally constant and compactly supported.

Let $\mathcal{S}(F)$ denote the $\mathbb{C}$-vector space of all Schwartz-Bruhat functions on F.

Definition 1.3. A LCA group is a locally compact hausdorff abelian group.

1.1 Additive Character

Let F be a local field, then in each case we can know that $(F, +)$ is a locally compact abelian group, which induces a Haar measure μ on F, then for any non-zero element $a \in F$, we can define a new measure on F by pull back:

$$\forall E \subset F \text{ measurable }, \quad \mu_a(E) = \mu(aE)$$

and we can conclude the following result:

Lemma 1.1. μ_a is a Haar measure on $(F, +)$, and there exists a constant factor $|a|$ such that $\mu_a = |a|\mu$, such a factor is independent of the choice of μ with

$$|a| = \begin{cases} \text{the standard absolute value if } F = \mathbb{R} \\ \text{the square of standard absolute value if } F = \mathbb{C} \\ N(\mathfrak{p})^{-v_\mathfrak{p}(a)} \text{ if } F \text{ is non-archimedean} \end{cases}$$

Proof. The case of $F = \mathbb{R}$ is clear. If $F = \mathbb{C}$, we take $a = u + iv$ and μ be the Lebesgue measure, let $z = x+iy$ and then $az = (ux-vy)+i(vx+uy)$, and we consider the integration in $\mathbb{R}^2$, then the Jacobian of the transformation is

$$|\det J_a| = \begin{vmatrix} u & -v \\ v & u \end{vmatrix} = u^2 + v^2 = \|a\|^2$$

so that shows $d(az) = \|a\|^2 dz$. If F is non-archimedean, we take μ be the Haar measure such that $\mu(\mathcal{O}) = 1$, and we set $q = N(\mathfrak{p}) = \#(\mathcal{O}/\mathfrak{p})$, then

$$1 = \mu(\mathcal{O}) = \mu(\bigsqcup_{i=1}^{q} x_i + \mathfrak{p}) = \sum_{i=1}^{q} \mu(x_i + \mathfrak{p}) = q\mu(\mathfrak{p})$$

so it shows that $\mu(\mathfrak{p}) = 1/q$, and same method shows that $\mu(\mathfrak{p}^n) = 1/q^n$. So for any $a \in F^\times$, we can write $a = \varpi^n u$ with $n = v_\mathfrak{p}(a)$ and $u \in \mathcal{O}^\times$, then $a\mathcal{O} = \varpi^n\mathcal{O} = \mathfrak{p}^n$ and then

$$\mu(a\mathcal{O})/\mu(\mathcal{O}) = 1/q^n = N(\mathfrak{p})^{-v_\mathfrak{p}(a)}$$

□

The constant factor can be extended to the whole field F by setting $|0| = 0$, and we call it **(normalized) module** on F. A obeservation here is that module is a power of standard absolute value on F, so $a \mapsto |a|$ defines a a continous map from F to $[0, \infty)$.

Remark. More generally, for any LCA group $(G, +)$ with Haar measure μ, if φ is a continuous automorphism of G, then $\mu \circ \varphi$ is also a Haar measure with $\mu \circ \varphi = \mathrm{mod}_G(\varphi) \cdot \mu$ for some constant factor only depending on φ, such a factor is called the module of φ on G. Let $\mathrm{Aut}(G)$ denote the group of all continuous automorphisms of G, then $\varphi \mapsto \mathrm{mod}_G(\varphi)$ is actually a homomorphism from $\mathrm{Aut}(G)$ to $\mathbb{R}_{>0}$. In the case of local field F, $F^\times$ can be embedded into $\mathrm{Aut}(F)$ by $a \mapsto m_a$ with $m_a(x) = ax$, and the module $\mathrm{mod}_F(m_a)$ is exactly $|a|$ defined in above lemma.

Definition 1.4. Additive character of a local field F is a continous group homomorphism $\psi : (F, +) \to \mathbb{S}^1$, and $\hat{F}$ is the set of all additive characters of F, it is called the dual group of F.

The dual group is indeed a group under pointwise multiplication, for topology on $\hat{F}$, we consider the compact-open topology, i.e. the topology generated by

$$[K, U] = \{\chi \in \hat{F} : \chi(K) \subset U\}$$

for every compact subset K of F and every open subset U of $\mathbb{S}^1$. It is a natural choice for functiona space, so dual group is ineed a topological group, a remarkable result here is that local field is self-dual.

Theorem 1.2. Let ψ be a non-trival additive character of F, then the map

$$\Psi : F \to \hat{F}, \quad a \mapsto \psi_a : x \mapsto \psi(ax)$$

is an isomorphism of LCA groups.

Proof. The proof depends on a fact that all local fields are a metrizable space such that compact subset is bounded. It is not hard to check that Ψ is a injective homomorphism, so firstly we prove that Ψ is continous: Let $[K, U]$ be an open basis of $\hat{F}$, then

$$\Psi^{-1}([K,U]) = \{a \in F : \psi_a(K) \subset U\} = \{a \in F : aK \subset \psi^{-1}(U)\}$$

notice that $\psi^{-1}(U)$ is a open since ψ is continous, so by continous map $m(x,y) = xy$ on $F \times F$, for any $y \in K$ there exists an neiborhood A_y of y and a neiborhood B_y of a such that $B_y A_y \subset \psi^{-1}(U)$. Let y runs over all K, and by compactness of K, there exists a finite index set I such that $K \subset \cup_{i\in I} A_i$, and we take $V_a = \cap_{i\in I} B_i$ as a neiborhood of a such that $V_a \subset \Psi^{-1}[K,U]$, it shows that a is an interior and then $\Psi^{-1}[K,U]$ is open.

Secondly, we prove that Ψ is an homemorphism to its image, it is enough to prove the image of open ball B_ϵ centred at 0 with radius ϵ is open in $\hat{F}$. Non-trival character ψ implies an element $b \in F$ such that $\psi(b) \neq 1$, and we construct a neiborhood of trivial character by $[\overline{B_R(0)}, W] \cap \Psi(F)$, where W is an neiborhood of 0 not cotaining $\psi(b)$, and $R \geq \|b\|/\epsilon$, then this neiborhood is contained in $\Psi(B_\epsilon)$. For other non-trival character in image, a translation of group allows us to conclude the result.

Finally, $\Psi(F)$ is complete since F is complete, and it is closed since $\hat{F}$ is hausdorff, so it is enough to prove that $\Psi(F)$ is dense in $\hat{F}$, that will imply $\Psi(F) = \hat{F}$. In fact, $x \in \Psi(F)^\perp$ implies that $\Psi(ax) = 1$ for any $a \in F$, and it shows that the annihilator is zero, so $\Psi(F)$ is dense in $\hat{F}$. ☐

Remark. A LCA group is not necessary self-dual, for example the dual group of $\mathbb{Z}$ is $\mathbb{S}^1$, and the dual group of $\mathbb{S}^1$ is $\mathbb{Z}$.

By the theory of harmonic analysis on LCA, if LCA group G is self-dual, then it is nice to identify the fourier tranform of a function f on G as a function on G itself, so it motivates us to define the transform on local field as follows:

Definition 1.5. Fix a non-trival additive character ψ and a Haar measure μ on F, the Fourier transform of type (ψ, μ) of a function $f \in L^1(F)$ is defined as

$$\hat{f}(y) = \int_F f(x)\psi(xy)\, d\mu(x)$$

There is no confusion to identify y as ψ_y by isomorphism, and the norm of ψ is always 1, so $|\hat{f}(y)| \leq \|f\|_{L^1} < \infty$ ensures that the transform is well-defined.

Definition 1.6. The standard additive character of F is defined as following:

- If $F = \mathbb{R}$, let $\psi(x) = e^{-2\pi i x}$;
- If $F = \mathbb{Q}_p$, let $\psi(x) = e^{2\pi i \{x\}}$, where $\{x\}$ is the fractional part of x defined by

  $$x = \sum_{n \geq N} a_n p^n \implies \{x\} = \sum_{n=N}^{-1} a_n p^n$$

  where $\{x\} = 0$ if $N \geq 0$, i.e. $x \in \mathbb{Z}_p$.

- Let $F_0 = \mathbb{R}$ or $\mathbb{Q}_p$, and F be a finite extension of F_0, let the standard additive character of F be the compistion
$$F \xrightarrow{\mathrm{Tr}_{F/F_0}} F_0 \xrightarrow{\psi_0} \mathbb{S}^1$$
where ψ_0 is the standard additive character of F_0.

In particular, if $F = \mathbb{Q}$, the standard character is $\psi(z) = e^{-2\pi i(z+\bar{z})}$. In the case of $F = \mathbb{Q}_p$, we have $\{x\} \in \mathbb{Z}[1/p]$ and $x - \{x\} \in \mathbb{Z}_p$, the intersection $\mathbb{Z}_p \cap \mathbb{Z}[1/p] = \mathbb{Z}$ implies that standard character ψ is a non-trival homomoprhism. In the aspect of continuity, we notice that $\mathbb{Z}_p \subset \ker \psi$ as an open subset containing 0, which shows that ψ is continous at 0, then ψ is continous on $\mathbb{Q}_p$ since it is a topological group. Hence ψ is exactly a non-trival additive character of $\mathbb{Q}_p$, It is not clear than the case $F = \mathbb{R}$.

From now on, we will always use standard character in the Fourier transform, so the transform will depend only on the choice of Haar measure, so a normalization can be done to obtain the inversion formula.

Proposition 1.3. Let F be a local field, then there exists a unique self-dual Haar measure μ, that means if $f \in \mathcal{S}(F)$, then $\hat{f} \in \mathcal{S}(F)$ and $\hat{\hat{f}}(x) = f(-x)$, explicitly the measure is given as following:

$$dx = \begin{cases} \text{lebesgue measure of } \mathbb{R} & \text{If } F = \mathbb{R} \\ 2 \times \text{lebesgue measure of } \mathbb{C} & \text{If } F = \mathbb{C} \\ \text{the Haar measure such that } \mu(\mathcal{O}) = N(\mathfrak{d})^{-\frac{1}{2}} & \text{If } F \text{ is non-archimedean} \end{cases}$$

Proof. For any Haar measure μ, $\hat{\hat{f}}(x) = cf(-x)$ holds for some constant c, so it is enough to choose a test function to modify. If $F = \mathbb{R}$, we choose $f(x) = e^{-\pi x^2}$ and calculate the transform under the lebesgue measure, than $\hat{f} = f$; If $F = \mathbb{C}$ we choose $f(x)$, and calculate the transform under $c \cdot dl$ (c times the lebesgue measure), than $\hat{f} = \frac{c}{2} f$ shows $2\,dl$ is the self-dual measure; If F is non-archimdean we choose $f = \mathbf{1}_{\mathcal{O}}$, then under some haar measure μ on a $\hat{\hat{\mathbf{1}}}_{\mathcal{O}} = \mu(\mathcal{O})\mu(\mathcal{O}^{\perp})\mathbf{1}_{\mathcal{O}}$, and we assume difference $\mathfrak{d} = \mathfrak{p}^m$ to calculate

$$\mu(\mathcal{O}^{\perp}) = \mu(\mathfrak{d}^{-1}) = \mu(\varpi^{-m}\mathcal{O}) = |\varpi|^{-m} N(\mathcal{O}) = N(\mathfrak{d})\mu(\mathcal{O})$$

so normization $\mu(\mathcal{O})\mu(\mathcal{O}^{\perp}) = 1$ gives the Haar measure. □

1.2 Multiplicative Character

The module on local field F induced a continous homomorphism

$$F^{\times} \to \mathbb{R}_{>0}, \quad x \mapsto |x|$$

the kernel of this homomorphism will be important in the following discussion, and we denote it by U, and we can conclude

$$U = \begin{cases} \{\pm 1\} & \text{If } F = \mathbb{R} \\ \mathbb{S}^1 & \text{If } F = \mathbb{C} \\ \mathcal{O}^{\times} & \text{If } F \text{ is non-archimedean} \end{cases}$$

The group U as a subspace of local field F is compact in each case, and also open if the F is non-archimedean.

Definition 1.7. a (multiplicative) quasi-character is a continous group homomorphism $\chi : F^\times \to \mathbb{C}^\times$, and the set of all quasi-characcters is denoted by $\mathcal{X}(F^\times)$.

a quasi-character χ is called a (unitary) character if $\mathrm{im}(\chi) \subset \mathbb{S}^1$; a quasi-character χ is called unramified if χ is trival on U.

The vocabulary here follows Tate's thesis, here are the examples in the case of archimedean local field, which is the classical result in harmonic analysis.

Example 1.4. Every quasi-character of $\mathbb{R}^\times$ is of the form

$$\chi_{\epsilon,s}(t) = \mathrm{sign}(t)^\epsilon \cdot |t|^s, \quad \epsilon \in \{0,1\}, s \in \mathbb{C}$$

Every quasi-character of $\mathbb{C}^\times$ is of the form

$$\chi_{n,s}(z) = (\frac{z}{\|z\|})^n \cdot |z|^s, \quad n \in \mathbb{Z}, s \in \mathbb{C}$$

where $\|z\|$ is the standard absolute value on $\mathbb{C}$, and in both case, the character is unitary if and only if s is a purely imaginary number.

By the example, it is natural to separate the quasi-character into two parts, the first part is a unitary character on units, and the second part is a power of module, in fact it is unramified and we can generalize the result.

Lemma 1.5. A quasi-character $\chi \in \mathcal{X}(F^\times)$ is unramified if and only if it is of the form $\chi(x) = |x|^s$ for some $s \in \mathbb{C}$. Let $\mathcal{U}(F^\times)$ denote the set of all unramified quasi-characters on $F^\times$, then it forms a subgroup of $\mathcal{X}(F^\times)$ with isomorphism

$$\mathcal{U}(F^\times) \cong \begin{cases} \mathbb{C} & \text{If } F \text{ is archimedean} \\ \mathbb{C}/\frac{2\pi i}{\log N(\mathfrak{p})}\mathbb{Z} & \text{If } F \text{ is non-archimedean} \end{cases}$$

Proof. Notice that χ is unramified if and only if χ factors through $|F^\times|$, because $U \subset \ker \chi$ if χ is trivial on U, while U is the kernel of the surjection $x \mapsto |x|$, that means there exists a continous homomorphism $\tilde{\chi} : |F^\times| \to \mathbb{C}^\times$ such that the following diagram commutes:

$$\begin{tikzcd} F^\times \arrow[r, "\chi"] \arrow[d, "x \mapsto |x|"'] & \mathbb{C}^\times \\ |F^\times| \arrow[ur, "\tilde{\chi}"'] & \end{tikzcd}$$

Hence if χ is unramified, it is enough to prove that continous homomorphism $\tilde{\chi} : |F^\times| \to \mathbb{C}^\times$ is of the form $x \mapsto x^s$ for some $s \in \mathbb{C}$. If F is archimedean, it is immediate by the example above, if F is non-archimedean, $|F^\times| = q^{\mathbb{Z}}$ with $q = N(\mathfrak{p})$, so the morphism is determined by the image of q, if we set $\tilde{\chi}(q) = a$, then we can choose $s = \log a$ with respect to a branch of logarithm.

Conversely, it is not hard to check that $\chi_s(x) = |x|^s$ is unramified for any $s \in \mathbb{C}$, which proves the first part of the lemma and induces a surjective map

$$\Phi : \mathbb{C} \to \mathcal{U}(F^\times), \quad s \mapsto \chi_s$$

and it is not hard to check $\mathcal{U}(F^\times)$ is indeed a subgroup of $\mathcal{X}(F^\times)$ and $\chi_{s+z} = \chi_s \cdot \chi_z$ for any $s, z \in \mathbb{C}$, which implies that Φ is a group homomorphism with kernel $\{s \in \mathbb{C} : |x|^s = 1, \forall x \in F^\times\}$. If F is archimedean, the kernel is trivial so Φ is an isomorphism; if F is non-archimedean, the kernel is exactly $\{s \in \mathbb{C} : |\varpi|^s = 1\} = \frac{2\pi i}{\log q}\mathbb{Z}$, so it induces a desired isomorphism. □

Lemma 1.6. The map $\chi \mapsto (\chi|_U, \chi \cdot (\chi|_U \circ \rho)^{-1})$ gives an isomorphism

$$\mathcal{X}(F^\times) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{cont}}(U, \mathbb{S}^1) \times \mathcal{U}(F^\times)$$

where $\rho : F^\times \to U$ is a retraction depending on the local field F, and every quasi-character χ is of the form $\chi = \tilde{\chi}|\cdot|^s$ for some $s \in \mathbb{C}$ and a unitary character $\tilde{\chi}$ conciding with χ on U.

Proof. It is enough to prove the non-arcimedean case, since we have seen that $\rho(x) = x/\|x\|$ when F is archimedean. Notice that $U = \mathcal{O}^\times$ is a compact subgroup of $F^\times$, then the restriction $\chi|_U$ must be a unitary, and we define $\rho(x) = \varpi^{-v_{\mathfrak{p}}(x)}x$, which is a continous homomorphism with $\rho(\varpi^n u) = u$ for any $n \in \mathbb{Z}$ and $u \in \mathcal{O}^\times$. Hence $c(x) = \chi \cdot (\chi|_U \circ \rho)^{-1}$ defines a quasi-character with $c(u) = 1$ for any $u \in \mathcal{O}^\times$, by lemma 1.5 it shows that c is unramified of the form $c(x) = |x|^s$ for some $s \in \mathbb{C}$, so $\chi = \chi_u|\cdot|^s$ with $\tilde{\chi} = \chi|_U \circ \rho$.

Conversely, we can construct a inverse map by $(c, \chi_s) \mapsto (c \circ \rho) \cdot \chi_s$, which proves the map is an isomorphism. □

Remark. The isomorphism is also topological, the retraction ρ induces a map

$$\rho^* : \mathrm{Hom}_{\mathrm{cont}}(U, \mathbb{C}^\times) \to \mathrm{Hom}_{\mathrm{cont}}(F^\times, \mathbb{C}^\times), \quad \chi \mapsto \chi \circ \rho$$

such a map is continous since local field is locally compact and Hausdorff, in this case the map defined in the lemma is a homeomorphism.

Lemma 1.7. Let $\chi : F^\times \to \mathbb{C}^\times$ be a quasi-character of a non-archimedean local field F, then there exists a integer $n \geq 0$ such that χ is trivial on $1 + \mathfrak{p}^n$.

Proof. $\mathbb{S}^1$ is a group has NSS (no small subgroup) properties, then there exists a neighborhood V of 1 in $\mathbb{S}^1$ such that V does not contain any non-trivial subgroup, then $\chi^{-1}(V)$ is a open neighborhood of 1 in $F^\times$. In fact, all open subgroups of $\mathcal{O}^\times$ form a chain as following:

$$\mathcal{O}^\times \supsetneqq 1 + \mathfrak{p} \supsetneqq 1 + \mathfrak{p}^2 \supsetneqq \cdots$$

and form a basis of open neighborhood of 1 in $F^\times$, so there exists a integer $n \geq 0$ such that $1 + \mathfrak{p}^n \subset \chi^{-1}(V)$, which implies that χ is trivial on $1 + \mathfrak{p}^n$. □

Hence there is no confusion to take the samllest integer $\mathfrak{f}$ and define it as the **conductor** of quasi-character χ, with a convention that $1 + \mathfrak{p}^0 = \mathcal{O}^\times$, then χ is unramified if and only if its conductor is 0, and such a concept measures how far a quasi-character is away from being unramified.

Remark. Let $F = \mathbb{Q}_p$ and $\chi : F^\times \to \mathbb{C}^\times$ be ramified character, and then $\chi|_{\mathbb{Z}_p^\times}$ factors through $\mathbb{Z}_p^\times/(1+p^{\mathfrak{f}}\mathbb{Z}_p)$ with conductor $\mathfrak{f} > 0$ as following

$$\mathbb{Z}_p^\times \longrightarrow \mathbb{Z}_p^\times/(1+p^{\mathfrak{f}}\mathbb{Z}_p) \xrightarrow{\sim} (\mathbb{Z}/p^n)^\times \xrightarrow{\varphi} \mathbb{S}^1$$

the proof is omitted here, and the interesting result is that the ramified part of χ is related to a primitive dirichlet character φ modulo $p^{\mathfrak{f}}$.

Later we will see that the local zeta integral will be defined on the character group, so it is better to treat character group as a complex manifold such that we can talk about the meromorphic continuation and functional equation.

Definition 1.8. Let χ and χ' be two quasi-character of $F^\times$, they are called equivalent if $\chi'|_U = \chi'|_U$, or equivalently, χ/χ' is unramified.

It defines a equivalence relation on $\mathcal{X}(F^\times)$, and the equivalence class of a quasi-character is $[\chi] = \{\chi|\cdot|^s : s \in \mathbb{C}\} = \chi \cdot \mathcal{U}(F^\times)$, and lemma 1.5 shows that $\mathcal{U}(F^\times)$ is indeed a complex lie group of dimension 1, so each class can be identified as a Riemann surface by translation of topological group, and thus $\mathcal{X}(F^\times)$ can be identified as a disjoint union of Riemann surfaces.

Finally, to consider the continuation of local zeta function, Tate introduces a notation to denote the "dual" of a quasi-character as following:

Definition 1.9. Let $\chi \in \mathcal{X}(F^\times)$, its **twisted dual** is defined as $\chi^\vee = \chi^{-1}|\cdot|$.

It is not hard to check that $\chi \mapsto \chi^\vee$ is a homeomorphism on $\mathcal{X}(F^\times)$ with $(\chi^\vee)^\vee = \chi$, and the image of a connected component $[\chi]$ is exactly the component $[\chi^{-1}]$.

1.3 Local Zeta Function

Now we will consider the integrand on $F^\times$, which will be used to define the local zeta integral.

Lemma 1.8. Let dx be the self-dual Haar measure on F defined in the previous, then

(a) $m(E) = \int_E \frac{dx}{|x|}$ defines a Haar measure on multiplicative group $F^\times$, and we denote it by $d^\times x$;

(b) If F is non-archimedean, $\mathcal{O}^\times$ gets measure $(1 - N(\mathfrak{p})^{-1})N(\mathfrak{d})^{-1/2}$.

(c) A function f is integrable on $F^\times$ with respect to $d^\times x$ if and only if $x \mapsto f(x)/|x|$ is integrable on $F - 0$ with respect to dx.

Proof. For any $a \in F^\times$, we have

$$\int_{aE} \frac{dx}{|x|} = \int_E \frac{d(ax)}{|ax|}$$

notice that $|x|$ is the module such that $d(ax) = |a|dx$ for any haar measure, which proves (a). For (b), we notice that $\mathcal{O}^\times = \mathcal{O} - \mathfrak{p}$, and in proof of lemma 1.1 we have seen that $\mathfrak{p}$ gets measure $N(\mathfrak{p})^{-1}$ with respect to $N(\mathfrak{d})dx$, so it is enough to calculate the result of (b). (c) is immediate by the definition of $d^\times x$. $\square$

Definition 1.10. Fix a function $f \in \mathcal{S}(F)$, the local zeta function associated with f is defined as a integrand on space $\mathcal{X}(F^\times)$ by

$$Z(f,\chi) = \int_{F^\times} f(x)\chi(x)\, d^\times x$$

Let χ be a quasi-character of the class $[\chi_0]$ of the form $\chi_0|\cdot|^s$, it is equivalent to write

$$Z(f,\chi_0,s) = \int_{F^\times} f(x)\chi_0(x)|x|^s d^\times x$$

and we call here s as the complex parameter of χ and $\sigma = \mathrm{Re}(s)$ as the **exponent** of χ.

Review that a function on Riemann surface $f : X \to \mathbb{C}$ is called holomorphic (resp. meomorphically) at $s \in X$ if there exists a local chart (U,ϕ) with $s \in U$ such that $f \circ \phi^{-1}$ is holomorphi (resp. meomorphically) at $\phi(s)$ as a complex function.

Lemma 1.9. Following the notation above, $\chi \mapsto Z(f,\chi)$ defines a holomorphic function on the domain of quasi-character with exponent $\sigma > 0$.

The lemma shows that the definition only makes sense on the half-plane of $\sigma > 0$.

Proof. We assume that K is a compact subset of the domain, for the archimedean case, schwartz condition implies that f is bounded by a constant B and f rapidly decreasing faster than any power of $1/|x|$ at infinity, so an estimate can be done by

$$|Z(f,\chi)| \le B\int_{0<|x|\le 1} |x|^{\sigma-1}\, dx + C_N \int_{|x|\ge 1} \frac{|x|^{\sigma-1}dx}{(1+|x|)^N}$$

$\sigma > 0$ ensures the first integral is finite, and the compactness of K ensures the existence of N such that $N > \sigma$ and the second integral is finite. If F is non-archimdean, suppose that $\mathrm{supp}(f) \subset \varpi^N\mathcal{O}$ for some integer $N \ge 0$, and then

$$|Z(f,\chi)| \le C\int_{0<|x|<N(\mathfrak{p})^N} |x|^\sigma d^\times x = C\sum_{n\ge -N}\int_{\varpi^n\mathcal{O}^\times} |x|^\sigma d^\times x$$

and lemma 1.8 (a) allows us to set $\mathrm{Vol}(\mathcal{O}^\times) = (1-N(\mathfrak{p})^{-1})N(\mathfrak{d})^{-1/2}$, then

$$|Z(f,\chi)| \le C\cdot \mathrm{Vol}(\mathcal{O}^\times)\sum_{n\ge -N} N(\mathfrak{p})^{-n\sigma} = C\cdot \mathrm{Vol}(\mathcal{O}^\times)\frac{N(\mathfrak{p})^{N\sigma}}{1-N(\mathfrak{p})^{-\sigma}}$$

the last equality holds when $\sigma > 0$ by geometric series. Finally, the inequalities above is enough to show the uniform convergence of $\chi \mapsto Z(f,\chi)$ in each compact of the component with $\sigma > 0$, so in each component the function is holomorphic on the domain $\sigma > 0$. $\square$

Lemma 1.10. Fix any two functions $f, g \in \mathcal{S}(F)$, and let χ be a quasi-character χ of exponent $0 < \sigma < 1$, then $\chi^\vee$ is of exponent $0 < \sigma < 1$ as well, and the following identity holds

$$Z(f,\chi)Z(\hat{g},\chi^\vee) = Z(\hat{f},\chi^\vee)Z(g,\chi)$$

Proof. Applying the Fubini's theorem to separate the integral

$$Z(f,\chi)Z(\hat{g},\chi^\vee) = \int_{(F^\times)} f(x)g(z)\chi(xy^{-1})\psi(yz)|yz|\, d^\times x\, d^\times y\, d^\times z$$

and then we change variable $y = xt$ and then $u = zt$ to calculate

$$\begin{aligned} Z(f,\chi)Z(\hat{g},\chi^\vee) &= \int_{(F^\times)^3} f(x)g(z)\chi(t^{-1})\psi(txz)|txz|\, d^\times x\, d^\times t\, d^\times z \\ &= (\int_{F^\times} g(z)\chi(z)\, d^\times z)(\int_{(F^\times)^2} f(x)\chi(u)^{-1}\psi(ux)|ux|\, d^\times x\, d^\times u) \\ &= Z(g,\chi)Z(\hat{f},\chi^\vee) \end{aligned}$$

notice that in finally equality we use the definition $|ux|d^\times x = |u|dx$ to recover the local zeta function. $\square$

Loosely speaking, the last lemma means the quotient of $Z(f,\chi)$ and $Z(\hat{f},\chi^\vee)$ is independent of the choice of function, so we can definie a meromorphic function on domain of quasi-character of exponent $0 < \sigma < 1$ by

$$\rho(\chi) = \frac{Z(\hat{f},\chi^\vee)}{Z(f,\chi)}$$

the choice of f should satisfy $\chi \mapsto Z(f,\chi)$ is not zero function, and we call it **ρ-factors** of local field F. The analytic continuation of ρ-factors will be the key to prove the local functional equation, Tate's strategy is to calculate each case explicitly.

Example 1.11. If $F = \mathbb{R}$, by example 1.4 we have

$$\mathcal{X}(\mathbb{R}^\times) = [\chi_0] \sqcup [\chi_1]$$

with χ_0 the trival character and χ_1 the sign character, and they are representative of even characters and odd characters respectively. In the component $[\chi_0]$, we can choose $f_0(x) = e^{-\pi x^2}$ with $\hat{f}_0 = f_0$; In the compoenent $[\chi_1]$, the choice of f_0 is not proper since its local zeta function will vanish, so we modify it by $f_1(x) = xe^{-\pi x^2}$ with $\hat{f}_1 = i\hat{f}_1$. After some calculation we can conclude the local zeta function

$$Z(f_\epsilon,\chi_{\epsilon,s}) = \pi^{-\frac{s+\epsilon}{2}}\Gamma\left(\frac{s+\epsilon}{2}\right)$$

and its ρ-factor

$$\rho(\chi_{\epsilon,s}) = i^\epsilon 2^{1-s}\pi^s \cos\frac{\pi(s+\epsilon)}{2}\,\Gamma(s)$$

for quasi-character $\chi_{\epsilon,s}(t) = \mathrm{sign}(t)^\epsilon|\cdot|^s$ of exponent $\sigma = \mathrm{Re}(s) \in (0,1)$.

Example 1.12. If $F = \mathbb{C}$, by example 1.4 we have

$$\mathcal{X}(\mathbb{C}^\times) = \bigsqcup_{n\in\mathbb{Z}}[c_n]$$

with $c_n(z) = (z/\|z\|)^n$ the unitary character, and we can choose schwartz functions f_n for each component $[c_n]$ by

$$f_n(z) = \begin{cases} (\bar{z})^n e^{-2\pi\|z\|^2} & n \geq 0 \\ z^{-n} e^{-2\pi\|z\|^2} & n < 0 \end{cases}$$

with fourier tranform $\hat{f}_n = i^{|n|} f_{-n}$, and in each case the local zeta function does not vanish, which allows us to calculate the local zeta function

$$Z(f_n, \chi_{n,s}) = (2\pi)^{1-s+\frac{|n|}{2}} \, \Gamma(s + \frac{|n|}{2})$$

and its ρ-factor

$$\rho(\chi_{n,s}) = (-i)^{|n|} (2\pi)^{1-2s} \frac{\Gamma(s + \frac{|n|}{2})}{\Gamma(1 - s + \frac{|n|}{2})}$$

for quasi-character $\chi_{n,s} = c_n |\cdot|^s$ of exponent $\sigma \in (0,1)$.

Example 1.13. If F is non-archimedean, let $\mathcal{X}_n(F^\times)$ denote the set of quasi-characters of conductor n, then we have a decomposition

$$\mathcal{X}(F^\times) = \bigsqcup_{n \geq 0} \mathcal{X}_n(F^\times)$$

where $\mathcal{X}_0(F^\times)$ is just the set of unramified quasi-characters. For each class $\mathcal{X}_n(F^\times)$, we can choose a test function $f_n \in \mathcal{S}(F)$ by

$$f_n(x) = \begin{cases} \psi(x) = \exp(2\pi i \, \{\mathrm{Tr}_{F/\mathbb{Q}_p}(x)\}) & \text{if } x \in \mathfrak{d}^{-1}\mathfrak{p}^{-n} \\ 0 & \text{otherwise} \end{cases}$$

and its transform are

$$\hat{f}_n(x) = \begin{cases} (N(\mathfrak{d})^{1/2} N(\mathfrak{p})^n) & \text{if } x \in 1 + \mathfrak{p}^n \\ 0 & \text{otherwise} \end{cases}$$

The ρ-factor of the unramified quasi-character $(n = 0)$ is

$$\rho(|\cdot|^s) = N(\mathfrak{d})^{s-1/2} \frac{1 - N(\mathfrak{p})^{s-1}}{1 - N(\mathfrak{p})^{-s}}$$

If c_n is a unitary character of conductor $n > 0$, then the ρ-factor of the ramified quasi-character is

$$\rho(c_n |\cdot|^s) = N(\mathfrak{d}\mathfrak{p}^n)^{s-\frac{1}{2}} W(\chi)$$

with root number $W(\chi)$ defined by

$$W(\chi) = N(\mathfrak{p})^{-n/2} \sum_{\bar{a} \in \mathcal{O}^\times/1+\mathfrak{p}^n} \chi(a) \psi(\frac{a}{\varpi^{d+n}}), \qquad \text{with } d = v_{\mathfrak{p}}(\mathfrak{d})$$

The case of unramified quasi-character is more simple, we give a explicit formula of local zeta function.

Lemma 1.14. If F is non-archimedean and χ is an unramified quasi-character, then

$$Z(\mathbf{1}_{\mathcal{O}}, \chi|\cdot|^s) = \frac{\mathrm{Vol}(\mathcal{O}^\times)}{1-\chi(\varpi)N(\mathfrak{p})^{-s}}$$

Proof. There eixsts a decomposition $F^\times = \sqcup_{n\in\mathbb{Z}}\varpi^n\mathcal{O}^\times$, and which allows us to calculate

$$\begin{aligned} Z(\mathbf{1}_{\mathcal{O}}, \chi|\cdot|^s) &= \sum_{n\geq 0}\int_{\varpi^n\mathcal{O}^\times}\chi(x)|x|^s d^\times x \\ &= \sum_{n\geq 0}\chi(\varpi)^n N(\mathfrak{p})^{-ns}\int_{\mathcal{O}^\times} d^\times x \\ &= \mathrm{Vol}(\mathcal{O}^\times)\sum_{n\geq 0}\chi(\varpi)^n N(\mathfrak{p})^{-ns} \end{aligned}$$

and the geometric series concludes the result. □

The calculations of other cases are omitted here, all of that can be found in Tate's thesis [1], or see the section 7.11 and 7.12 in [2]. The non-archimedean case is more complicated and more arithmetically meaningful, the root number $W(\chi)$ is of norm one and can be viewed as a generalization of Gauss sum. Moreover, the ρ-factors above has a non-vanishing meromorphic continuation to whole space $\mathcal{X}(F^\times)$, which is immediate by the explicit formula above, and this function is independent of the choice of test function, which in only depends on the quasi-character χ, standard additive character ψ and the additive Haar measure dx, the last two data are fixed in our context.

Theorem 1.15 (Tate's local functional equation)**.**
Let F be a local field and fix a function $f \in \mathcal{S}(F)$, then $\chi \mapsto Z(f,\chi)$ has a meromorphic continuation to whole space $\mathcal{X}(F^\times)$, and the following functional equation holds

$$Z(\hat{f}, \chi^\vee) = \rho(\chi)Z(f,\chi)$$

Proof. In each case of local field, $\rho(\chi) = Z(\hat{f_C}, \chi^\vee)/Z(f_C,\chi)$ with f_C choosen as above examples, so ρ has a meomorphic continuation to whole space $\mathcal{X}(F^\times)$. Let $h(\chi) = \rho(\chi)Z(f,\chi)$, lemma 1.9 shows that h is meomorphic on the domain of exponent $\sigma > 0$, and by lemma 1.10 we can calculate that $h(\chi) = Z(\hat{f},\chi^\vee)$ in the domain of exponent $0 < \sigma < 1$, and $\chi \mapsto Z(\hat{f},\chi^\vee)$ defines a holomorphic function on the domain of exponent $\sigma < 1$. Hence the continuation of $\chi \mapsto Z(f,\chi)$ can be concluded by the continuation of h by the uniqeness of meromorphic continuation. □

Example 1.16. Some calculation can be done for some sepecific example, we can consider a number field, its completion are r_1 real places and r_2 complex places, and the rest are the local field at some prime ideaal $\mathfrak{p}$. If $F = \mathbb{Q}$ and we choose $f(x) = e^{-\pi x^2}$, then we can calculate

$$Z(f,s) = 2\int_0^\infty e^{\pi x^2}x^{s-1}\,dx = \pi^{-s/2}\Gamma(s/2)$$

If $F = \mathbb{Q}_p$ and we choose $f(x) = \mathbf{1}_{\mathbb{Z}_p}$, then use the fact that $\mathbb{Z}_p - \{0\} = \bigcup_{n\geq 0} p^n\mathbb{Z}_p^\times$ we can calculate

$$\begin{aligned}
Z(f,s) &= \sum_{n\geq 0}\int_{p^n\mathbb{Z}_p^\times} |x|_p^s\, d^\times x \\
&= \sum_{n\geq 0} p^{-ns}\int_{\mathbb{Z}_p^\times} d^\times x \\
&= \sum_{n\geq 0} p^{-ns} = \frac{1}{1-p^{-s}}
\end{aligned}$$

Pay attention that in both case, the last equality holds when $\mathrm{Re}(s) > 0$, and by Euler's product formula we can find that the proudct of all local zeta functions is so-called completed zeta function

$$s \mapsto \pi^{-s/2}\Gamma(s/2)\zeta(s)$$

which plays an important role in the functional equation of Riemann zeta function, it can go back to Riemann's famous short paper [3].

In the example above, variable s can be seen as a parameter of unramified quasi-character $\chi_s = |\cdot|^s$, and beyond the example, we should consider other type of zeta function. For instance, a dirichlet L-function asspciated with a dirichlet character χ, which is defined by the following Euler product

$$L(s,\chi) = \prod_{p \text{ prime}} \frac{1}{1-\chi(p)p^{-s}}$$

it suggests that we should consider all type of quasi-character.

2 Global Functional Equation

2.1 Adele Ring

Firstly we introduce a construction of the topological space, which allows us to obtain a finer topology than the product topology.

Definition 2.1. Let $\{X_i\}_{i\in I}$ be a family of topological spaces, and $\{U_i\}_{i\in I}$ be a family of open subset of X_i, then the **restricted direct product of $\{X_i\}_{i\in I}$ with respect to** $\{U_i\}_{i\in I}$ is defined as a subspace of direct product $\prod_{i\in I} X_i$ as following

$$\prod_{i\in I}{}' (X_i, U_i) := \{(x_i)_{i\in I} \in \prod_{i\in I} \mid x_i \in U_i \text{ for almost all } i \in I\}$$

it can be viewed as a topological space with the open basis given by

$$\mathcal{B} = \{\prod_{i\in I} V_i \mid V_i = U_i \text{ for almost all } i \in I\}$$

If each X_i is a measurable space with σ-alegbra $\mathcal{A}_i$ and a measure μ_i, if $U_i \in \mathcal{A}_i$ for each i and $\mu_i(U_i) = 1$ for almost all i, then we can define a measure on the restricted direct product as following: Let $\mathcal{A} = \sigma(\mathcal{B})$ be the σ-algebra of restricted direct product, where $\mathcal{B}$

are set of elements of the form

$$M = \prod_{i \in I} M_i \quad M_i \in \mathcal{A}_i \text{ and } M_i = U_i \text{ for almost all } i \in I$$

then we define the μ on generated sets by

$$\mu(M) := \prod_{i \in I} \mu_i(M_i)$$

then μ can be extended to a measure by the Carathéodory extension theorem, and it is called the **restricted product measure** of $\{\mu_i\}_i$ with respect to $\{U_i\}_i$.

Lemma 2.1. Let $\{G_i\}_{i \in I}$ be a family of locally compact abelian groups, and $\{U_i\}_{i \in I}$ be a family of compact-open subgroups $U_i \subset G_i$, then

(a) the restricted direct product $G = \prod'_{i \in I}(G_i, U_i)$ is a locally compact abelian group under the pointwise operation.

(b) If μ_i is a Haar measure for each G_i and $\mu_i(U_i) = 1$ for almost all i, then the restricted product measure μ is a Haar measure on G.

Proof. G is clear an abelian group, and let $S \subset I$ be a finite subset and we define a subgroup G_S of G by

$$G_S = \prod_{i \in S} G_i \times \prod_{i \notin S} U_i$$

It is locally compact since first part is open and locally compact (finite direct product of locally compact group is locally compact) and the second part is compact (Tychonoff's theorem), so we can rewrite the G by

$$G = \bigcup_{S \subset I \text{ finite}} G_S$$

so it is enough to prove (a). In context of LCA groups, $\mathcal{A}$ can be viewed as the Borel σ-algebra, so we can verify (b) by checking the translation invariance of μ on the open basis $\mathcal{B}$. Let $g = (g_i)_i \in G$ and $M = \prod_{i \in I} M_i \in \mathcal{B}$, then

$$\mu(gM) = \prod_{i \in T \cup L} \mu_i(g_i M_i) = \prod_{i \in T \cup L} \mu_i(M_i) = \prod_{T} \mu_i(M_i) = \mu(M)$$

where T is choosen to be the finite subset of I such that $M_i = U_i$ if $i \not{} nT$, and L is the finite subset of I such that $g_i \notin U_i$ if $i \in L$. $\square$

Another useful lemma is a generalization of the Chinese remainder theorem, which allows us to find exactly one element in number field K which statisfy some local conditions at finite places.

Lemma 2.2 (Weak approximation theorem)**.** Given n inequivalent non-trival valuations $\| \cdot \|_i$ for $i = 1, \ldots, n$, then for any $(a_1, \ldots, a_n) \in F^n$ and $\epsilon > 0$, there exists an element $a \in F$ such that $\|a - a_i\|_i < \epsilon$ for any $i = 1, \ldots, n$.

A proof can be founded in theorem 4.1.3 of [2], we omit it here. Now with above lemmas, formal definition of adele ring can be given.

Definition 2.2. Let K be a number field, then the adele ring $\mathbb{A}$ of K is defined as the restricted direct product of $\{K_v\}_{v\in V(K)}$ with respect to $\{\mathcal{O}_v\}_{v\in V(K)}$, where $\mathcal{O}_v = K_v$ if v is archimedean, and $\mathcal{O}_v$ is the valuation ring of K_v if v is non-archimedean. The operation of addition and multiplication on $\mathbb{A}_K$ are defined pointwisely by

$$(ab)_v = a_v b_v, \quad (a+b)_v = a_v + b_v$$

For any $a, b \in \mathbb{A}_K$.

We can only consider the finite place of K, that means the restricted direct product of $\{K_v\}_{v|\infty}$ with repect to $\{\mathcal{O}_v\}_{v|\infty}$, which is denoted by $\mathbb{A}_f$ and call it **the finite adele ring** of K. Review that the minkowski space K_∞ of K is defined by the embedding of K into $\mathbb{R}^{[K:\mathbb{Q}]}$, and $K_\infty \cong K \otimes_{\mathbb{Q}} \mathbb{R} \cong \prod_{v\nmid\infty} K_v$, then the adele ring can be decomposed as the direct product

$$\mathbb{A}_K \cong K_\infty \times \mathbb{A}_{K,f} \cong K \otimes_{\mathbb{Q}} \mathbb{A}_{\mathbb{Q}}$$

Applying the part (a) of lemma 2.1 to finite adele part, then it is enough to conclude that $(\mathbb{A}, +)$ is a LCA group. In fact, $\mathbb{A}$ is a topological ring, and we stress $(\mathbb{A}, +)$ as adele group to distinguish.

Definition 2.3. The diagonal embedding of K into $\mathbb{A}$ is defined by

$$\iota : K \to \mathbb{A}, \quad a \mapsto (a, a, a, \dots)$$

It is not hard to check that ι is an injective ring homomorphism, so we identified K as the subgroup of adele group by the diagonal embedding, and the topology of K is just the subspace topology of $\mathbb{A}$.

Lemma 2.3. *K is discrete and $\mathbb{A}_K/K$ is compact under the quotient topology.*

Proof. For any archimedean place v take a open subset U_v to construct an open neighborhood of zero in $\mathbb{A}$ by

$$U = \prod_{v\nmid\infty} U_v \times \prod_{v|\infty} \mathcal{O}_v$$

Notice that if $K \cap U$ is non-empty, then any element in the intersection must be in $\mathcal{O}_K$ since the intersection of $\cap_{v|\infty}\mathcal{O}_v$ with K is exactly $\mathcal{O}_K$, that allows us to conclude

$$K \cap U \subset \mathcal{O}_K \cap \prod_{v|\infty} U_v$$

By Minkowski's embedding, $\mathcal{O}_K$ is a lattice in K_∞ then there exists U_v such that the intersection is zero adele element, which proves that zero is an isolated point in K, hence K is discrete. □

Lemma 2.4. *$\mathbb{A}_K/K$ is compact under the quotient topology.*

Proof. Let $B \subset K_\infty$ be the fundamental domain of $\mathcal{O}_K$ in K_∞, it is compact and allows us to construct a compact subset

$$W = B \times \prod_{v|\infty} \mathcal{O}_v \subset \mathbb{A}$$

then we claim that $\mathbb{A} = W + K$, then for canonical map $\pi : \mathbb{A} \to \mathbb{A}/K$ we can conclude that $\mathbb{A} = \pi(W)$ is compact by continuity.

To finish the proof of the claim, it is enough to prove $\mathbb{A} \subset K + W$. For any $a \in \mathbb{A}$, there exists a finite subset S of finite places such that $a_v \in \mathcal{O}_v$ for any $v \notin S$, then by weak approximation (lemma 2.2) we can finish the proof. $\square$

From now on, for any local field $K_v/\mathbb{Q}_{p_v}$, we fix dx_v to be the self-dual Haar measure in the context of lemma 1.3, notice that $K_v/\mathbb{Q}_{p_v}$ is an unramified extension for almost all finite place v, so $\mathcal{O}_v$ gets measure 1 for almost all finite places v, then lemma 2.1 implies a Haar measure on $\mathbb{A}$, and we denote it by $dx = \prod_v dx_v$.

Definition 2.4. Let $x = (x_v)_v \in \mathbb{A}$, then the adele norm of x is defined as

$$|x|_{\mathbb{A}} = \prod_v |x_v|_v$$

where $|\cdot|_v$ is the module (rf. lemma 1.1) on local field K_v of K.

Remark. For any x, there existsa finite subset S of places suhch that S contains all archimedean places and $x_v \in \mathcal{O}_v$ for any $v \notin S$, then

$$\|x\|_{\mathbb{A}} = \prod_{v\in S} |x_v|_v \cdot \prod_{v\notin S} |x_v|_v \leq \prod_{v\in S} |x_v|_v < \infty$$

which measn the adele norm is well-defined.

Lemma 2.5. For any $a \in K^\times$, we have $d(ax) = |a|_{\mathbb{A}} dx$.

Proof. For any $a \in K^\times$, we fix a finite subset S such that $|a_v| = 1$ if $v \notin S$, then we take a borel subset $E = \prod_v E_v$ with a finite subset T such that E_v gets measure 1 if $v \notin T$, then we have

$$\int_E d(ax) = \prod_{v\in S\cup T} \int_E d(a_v x_v) = \prod_{v\in S\cup T} |a_v|_v \int_E dx_v = \prod_{v\in S} |a_v|_v \prod_{v\in T} \int_E dx_v$$

then the result is immediate. $\square$

Remark. The automorphism of adele group can be given by the multiplication $x \mapsto ax$ if $a \in \mathbb{A}^\times$, it is similar to the local arithmetical field, and follow the remark of lemma 1.1, a conclusion is that the adele norm is the module of the automorphism of adele group given by multiplication, so the lemma holds true for any Haar measure on $\mathbb{A}$.

2.2 The Adelic harmonic analysis

This section is analogous to the section 1.2, that means the fourier analysis will be developed on the adele ring, so firstly the dual group structure should be determined.

Definition 2.5. Let $\{f_v\}_v$ be a family of functions $f_v : K_v \to \mathbb{C}^\times$ such that $f_v|_{\mathcal{O}_v} = 1$ for all but finitely many v, then for any $x = (x_v)_v \in \mathbb{A}_K$, a product of $\{f_v\}_v$ is defined as following

$$(\prod_v f_v)(x) := \prod_v f_v(x_v)$$

The condition on the family of functions is necessary to ensure the product is finite, so the product defines a function on $\mathbb{A}_K$.

Definition 2.6. A Schwartz-Bruhat function on $\mathbb{A}_K$ is defined as the product $\prod_v f_v$, with each $f_v \in \mathcal{S}(K_v)$ and $f_v = \mathbf{1}_{\mathcal{O}_v}$ for almost all finite place v. And the space of Schwartz-Bruhat functions on $\mathbb{A}_K$ is denoted by $\mathcal{S}(\mathbb{A}_K)$.

From now on, we fixe ψ_v to be the standard character on K_v in the context of definition 1.6, and $\psi = \prod_v \psi_v$. As we have seen in the section 1.2, each local field K_v is self-dual, and the dual group of $K_v/\mathcal{O}_v$ is $\mathcal{O}_v$ if v is non-archimedean, then an similar consequence is that $\mathbb{A}_K$ is self-dual.

Theorem 2.6. The map

$$\Psi : \mathbb{A}_K \to \widehat{\mathbb{A}_K}, \quad a \mapsto \psi(a-) = \prod_v \psi_v(a_v-)$$

gives an isomorphism of LCA groups.

Proof. Each $\psi_v(a_v-)$ gives an isomophism between K_v and $\hat{K_v}$ by the theorem 1.2, and for almost all place$\Psi_v(a_v-)$ is trival on $\mathcal{O}_v$, so that means the map Ψ gives an isomorphism to restricted direct product $\prod_v'(\hat{K_v}, \widehat{K_v/\mathcal{O}_v})$, noticed that $\widehat{K_v/\mathcal{O}_v}$ can be identified as $\mathcal{O}_v$ by duality of LCA group we introduced in section 1.2, so we can conclude the result. ☐

Corollary 2.7. $\hat{K}$ is isomorphic to $\mathbb{A}_K/K$ as LCA groups.

The analogue of $\mathbb{Z} \subset \mathbb{R}$ and $K \subset \mathbb{A}_K$ allows us to develop the adelic harmonic analysis.

Definition 2.7. Let $f \in \mathcal{S}(\mathbb{A}_K)$, the fourier transform of f is defined as integrand

$$\hat{f}(y) := \int_{\mathbb{A}_K} f(x)\psi(xy)\, dx$$

where $dx = \prod_v dx_v$ is the product of self-dual Haar measure in the context of proposition 1.3.

Notice that $K_v/\mathbb{Q}_{p_v}$ is unramified for almost all finite place v, so $\mathcal{O}_v$ gets measure $N(\mathfrak{d}_v)^{-1/2} = N(\mathcal{O}_v)^{-1/2} = 1$ for almost all finite places v, hence dx here is exactly a Haar measure by lemma 2.1. In fact, the measure dx choosen here is self-dual, and it is enough to choose a nice function $f \in S(\mathbb{A}_K)$ to verify.

Review that K is a discrete subgroup of $\mathbb{A}$, and its dual group is a compact group $\mathbb{A}/K$, this is analogous to the classical duality between $\mathbb{Z}$ and $\mathbb{S}^1$, so the fourier series can be developed similarly.

Theorem 2.8 (Quotient integral formula)**.** Let G be a LCA group and H its closed subgroup, then for any two Haar measure μ_G and μ_H on G and H respectively, there exists a unique Haar measure $\mu_{G/H}$ on the quotient group G/H such that for any $f \in L^1(G)$, we have

$$\int_G f(g)d\mu_G(g) = \int_{G/H} \left(\int_H f(gh)d\mu_H(h) \right) d\mu_{G/H}(gH)$$

A proof can be found in theorem 1.5.3 of [4], this theorem allows us to decompose the integration. By the quotient integral formula, the Haar measure dx on $\mathbb{A}$ and counting measure on K induces a Haar measure on $\mathbb{A}/K$, we denote it by $d\bar{x}$.

Definition 2.8. A function f on $\mathbb{A}_K$ is called K-periodic if $f(x+a) = f(x)$ for all $x \in \mathbb{A}_K$ and $a \in K$. Let $f \in L^1(\mathbb{A}/K)$, its fourier transform is defined as an integrand on K by

$$\hat{f}(y) := \int_{\mathbb{A}/K} f(x)\psi(xy)\, d\bar{x}$$

If we define the fundamental domain of $\mathbb{A}/K$ as D, then equivalently, the transform can be defined by

$$\hat{f}(y) = \int_D f(x)\psi(xy)\, dx$$

then the fourier series of f can be given by inversion formula

$$f(x) \approx \frac{1}{\mathrm{Vol}(D)} \sum_{a \in K} \hat{f}(a)\overline{\psi(ax)}$$

K is discrete then the integrable condition on K will reduce to the sum condition, so if $\hat{f} \in l^1(K)$ i.e. $\sum_{a\in K} |\hat{f}(a)| < \infty$, then the fourier series converges uniformly to f.

Theorem 2.9 (Poisson summation formula)**.** If $f \in \mathcal{S}(\mathbb{A})$, then

$$\sum_{a \in K} f(a) = \sum_{a \in K} \hat{f}(a)$$

Proof. Take a function on $\mathbb{A}$ by $F(x) = \sum_{a\in K} f(a+x)$, and we claim that $F \in L^1(\mathbb{A}/K)$ to consider its fourier transform

$$\begin{aligned}\hat{F}(k) &= \int_D \sum_{a\in K} f(a+x)\psi(kx)dx \\ &= \sum_{a\in K} \int_{a+D} f(z)\psi(kz)dz \\ &= \int_{\mathbb{A}} f(z)\psi(kz)dz = \hat{f}(k)\end{aligned}$$

for equality $(*)$, we change variable by $z = a+x$ and use the facts that additive character ψ is trivial on K. Hence it allows us to get the fourier series of F

$$F(x) = \frac{1}{\mathrm{Vol}(D)} \sum_{a+K} \hat{f}(a)\overline{\psi(ax)}$$

while the series here is exactly convergent to F since $\hat{f} \in l^1(K)$ by

$$\sum_{a\in K} |\hat{f}(a)| \le \int_{\mathbb{A}} |\hat{f}(z)|dz < \infty$$

so take $x = 0$ we can conclude that

$$\sum_{a\in K} f(a) = \frac{1}{\mathrm{Vol}(D)} \sum_{a\in K} \hat{f}(a)$$

while $\hat{f} \in \mathcal{S}(\mathbb{A})$ means that same argument can be done to obtain

$$\sum_{a \in K} \hat{f}(a) = \frac{1}{\mathrm{Vol}(D)} \sum_{a \in K} f(a)$$

By comparison $\mathrm{Vol}(D) = 1$ and the poisson's summation is immedeiately obtained.

Finally, we finish the proof of claim that $F \in L^1(\mathbb{A}/K)$. It is enough to prove that the sum converges uniformly, because it shows that F is a K-periodic continous function on $\mathbb{A}$. Given $f = \prod_v f_v$ with each $f_v \in \mathcal{S}(K_v)$ and $f_v = \mathbf{1}_{\mathcal{O}_v}$ for almost all finite place v to rewrite F

$$F(x) = \sum_{a \in K} \prod_v f_v(x_v + a)$$

and it is enough to consider the compact $C = \prod_v C_v$ with each C_v is a compact subset of K_v and $C_v = \mathcal{O}_v$ for almost all finite place v. Firstly for finite places, $f_v(x_v + a) = 0$ if and only if a is in a compact $M_v = \mathrm{supp}(f_v) - S_v$, the the compactness allows us to find a integer N_v such that we can construct

$$I = \{a \in K : |a|_v \leq N(\mathfrak{p})^{N_v}, v \mid \infty\}$$

and $\prod_{v|\infty} f_v(x_v + a)$ supports on I. While $N_v = 0$ for almost all v since $M_v = \mathcal{O}_v$ for almost all v, then I is exactly a fractional ideal of K. Then an estimation can be done by

$$|F(x)|_{\mathbb{A}} \leq \sum_{a \in I} \prod_{v \nmid \infty} |f_v(a + x_v)|_v \prod_{v | \infty} |f_v(a + x_v)|_v \leq B \sum_{a \in I} \prod_{v \nmid \infty} |f_v(a + x_v)|_v$$

where $B = \sup_{a \in I} \prod_{v|\infty} |f_v(a + x_v)|_v$ is finite, and then I is exactly a lattice in K_∞ by minkowski's embedding, so the sum on the right hand side is finite by finite-dimensional fourier analysis, which proves the uniform convergence. □

Another consequence in the proof is that $\mathrm{Vol}(D) = 1$, which allows us to prove a famous result in a more elegant way.

Corollary 2.10 (Product formula)**.** For any $a \in K^\times$, we have $|a|_{\mathbb{A}} = 1$.

Proof. By module property of adele norm we have $\mathrm{Vol}(aD) = |a|_{\mathbb{A}}\mathrm{Vol}(D)$. While D is a fundamental domain and $aK = K$, then aD is also a fundamental domain, that means $\mathrm{Vol}(aD) = \mathrm{Vol}(D)$, so the result is obtained since $\mathrm{Vol}(D) = 1$. □

2.3 Idele Group and Hecke Character

The unit group of the adele ring $\mathbb{A}_K^\times$ is not a topological group (the subspace topology of $\mathbb{A}_K$) under the multiplication, the reason is that the inverse map $(a_v)_v \mapsto (a_v^{-1})_v$ is not continous, so a finer topology should be given to the unit group for a further study.

Definition 2.9. The idele group of K is defined as the restricted direct product of $K_v^\times$ with respect to $\mathcal{O}_v^\times$, where $\mathcal{O}_v^\times = K_v^\times$ if v is archimedean, and $\mathcal{O}_v^\times$ is the unit group of $\mathcal{O}_v$ if v is non-archimedean.

Remark. It is a multiplicative locally compact abelian group, and analogously to the adele ring, **the finite idele group** of K is defined as the restricted direct product of $\{K_v^\times\}_{v\nmid\infty}$

with respect to $\{\mathcal{O}_v^\times\}_{v\nmid\infty}$ for all non-archimedean place v, which is denoted by $\mathbb{A}_{K,f}^\times$, and idele group can be decomposed as the direct product

$$\mathbb{A}_K^\times \cong K_\infty^\times \times A_{K,f}^\times$$

Lemma 2.11. $|\cdot|_{\mathbb{A}} : \mathbb{A}^\times \to \mathbb{R}_{>0}$ is a continous and surjective homomorphism.

Proof. The multiplicative property is clear by definition. For continuity it is enough to check at the identity $(1, 1, \dots)$, and we take a voisiange

$$V = \prod_{v\nmid\infty} U_v \times \prod_{v|\infty} \mathcal{O}_v^\times$$

then $||x|_{\mathbb{A}} - 1| = |\prod_{v\nmid\infty} |x|_v - 1|$. Notice that $x \mapsto |x|_v$ is continous for each v, which allows us to define a continous function $\varphi(x) = \prod_{v\nmid\infty} |x|_v$ on $\prod_{v\nmid\infty} K_v$, then the continuity of $(1, ..., 1)$ finishes the proof. For surjective we fix a real place u to construct an element $x \in \mathbb{A}$ with $x_v = 1$ for any $v \neq u$, then $|x|_{\mathbb{A}} = |x_u|_{\mathbb{R}}$, so the surjectivity is obtained by the surjectivity of $|\cdot|_{\mathbb{R}}$. $\square$

Remark. Adele norm is not continuous on $\mathbb{A}$, this is a more delicated point why the topology of idele group is finer. To see this, we can take a sequence of adele elements x_n by $(x_n)_v = 0$ if v is the n-th place, and $(x_n)_v = 1$ for any other place v, then x_n converges to $\mathbf{1} = (1, \dots, 1)$, if the adele norm is continuous, then $|x_n|_{\mathbb{A}}$ should converge to $|\mathbf{1}|_{\mathbb{A}} = 1$, but $|x_n|_{\mathbb{A}} = 0$ for any n, which is a contradiction.

The kernel of the norm is defined as norm-one idele group

$$\mathbb{A}_1^\times = \{x \in \mathbb{A} : |x|_{\mathbb{A}} = 1\}$$

It is a closed subgroup of idele group. Similarly we restrict the digonal embedding of K into $K^\times$, the finer topology allows us to identify $K^\times$ as a discrete subggroup of idele gruop, while the quotient group $\mathbb{A}^\times/K^\times$ is not compact, otherwise the induced map $\mathbb{A}^\times/K^\times \to \mathbb{R}_{>0}$ by adele norm is surjective and continous, which makes a contradiction that the image $\mathbb{R}_{>0}$ is compact.

Lemma 2.12. $A_1^\times/K^\times$ is compact under the quotient topology.

A proof can be found in 7.4.2 of [2], it should apply the argument about dirichlet unit theorem. We define the idele class group of K as $C_K = \mathbb{A}^\times/K$, and the norm-one idele class group of K as $C_K^1 = \mathbb{A}_1^\times/K^\times$, the adele norm on idele group induces a exact sequence of LCA groups

$$1 \to C_K^1 \to C_K \xrightarrow{|\cdot|_{\mathbb{A}}} \mathbb{R}_{>0} \to 1$$

In short words, the idele class group is difficult to study since it is a generalization of the ideal class group, and we will interested in the character on the class group.

Lemma 2.13. Let $\chi : \mathbb{A}^\times \to \mathbb{C}^\times$ be a continous homomorphism, define the local quasi-character at each place v by

$$\chi_v : K_v \to \mathbb{C}^\times, \quad a_v \mapsto \chi(1, \dots, \underbrace{a_v}_{v \text{ place}}, \dots, 1)$$

then $\chi = \prod_v \chi_v$ and there eixists a finite subset S of places, it contains all archimdean places and χ_v is unramified for any $v \notin S$.

Proof. The product $\prod_v \chi_v$ is always finite by the definition of idele group, and an algebraic calculation shows it is equal to χ. Let $\mathbb{A}_f^\times$ be the finite idele group, and then consider the restriction of χ on $\mathbb{A}_f^\times$, and denote it by χ_f. Finite idele group is a totally disconnected LCA group, so by NSS property of $\mathbb{C}^\times$, we can conclude a open neighborhood U of identity in $\mathbb{A}_f^\times$ such that χ_f is trival on U, and then the topology of $\mathbb{A}_f^\times$ implies the existence of a finite subset S of places such that

$$\prod_{v \in S} U_v \times \prod_{v \notin S} \mathcal{O}_v^\times \subset U, \quad v \text{ finite}$$

where U_v is an open subgroup of $K_v^\times$. Adding all archimedean places into S, we can conclude that χ_v is unramified for any $v \notin S$. $\square$

As local multiplicative characters defined in the section 1.3, we shall also develop the global multiplicative characters such that we can define the global zeta function.

Definition 2.10. A **Hecke character** or a **quasi-character** of K is a continous homomorphism $\chi : \mathbb{A}_K^\times \to \mathbb{C}^\times$ such that $\chi|_{K^\times} = 1$, and its twisted dual is defined by $\chi^\vee = \chi^{-1}| \cdot |_{\mathbb{A}}$.

Remark. Equivalently, a Hecke character can be defined as a continous homomorphism on idele class group C_K.

By product formula (corollary 2.10) and lemma 2.11, $| \cdot |_{\mathbb{A}}^s$ defines a Hecke character for any $s \in \mathbb{C}$, and there are a condition to determine whether a Hecke character is of this form, which is analogous to lemma 1.5 to determine unramified local quasi-character.

Lemma 2.14. A Hecke character $\chi : \mathbb{A}^\times \to \mathbb{S}^1$ is of the form $| \cdot |_{\mathbb{A}}^s$ for some $s \in \mathbb{C}$ if and only if χ is trivial on subgroup $\mathbb{A}_1^\times$.

Proof. $\mathbb{A}_1^\times$ is the kernel of $| \cdot |_{\mathbb{A}}$, so it is enough to prove the necessity. If $\mathbb{A}_1^\times \subset \ker \chi$, then χ factors through the quotient group $\mathbb{A}^\times / \mathbb{A}_1^\times$ such that a diagram commutes

$$\begin{tikzcd}
\mathbb{A}^\times \arrow[r, "\chi"] \arrow[d] & \mathbb{C}^\times \\
\mathbb{R}_{>0} \cong \mathbb{A}^\times / \mathbb{A}_1^\times \arrow[ur, dotted, "\bar{\chi}"'] &
\end{tikzcd}$$

then the result is immediate by the quasi-character of $\mathbb{R}_{>0}$. $\square$

Lemma 2.15. For any Hecke character χ, there exists a complex number s and a unitary Hecke character η such that $\chi = \eta | \cdot |_{\mathbb{A}}^s$.

Proof. It is enough to prove that $|\chi| = | \cdot |^s$ for some $s \in \mathbb{C}$, then the result is obtained by taking $\eta = \chi |\chi|^{-1}$. For any $x \in K^\times$, we have $|\chi(x)| = |1| = 1$, so $|\chi|$ is eaxctly a Hecke character, and we veiw it as a character on idele class group C_K to $\mathbb{R}_{>0}$. While C_K^1 is compact, so image of C_K^1 under $|\chi|$ is compact, then the image must be trival by the structure of $\mathbb{R}_{>0}$, so $|\chi|$ is trivial on C_K^1, by lemma 2.14 it is of the form $| \cdot |_{\mathbb{A}}^s$ for some $s \in \mathbb{C}$. $\square$

Similarly, two Hecke chracters χ and χ' on class group C_K are called equivalent if they concide on the compact subgroup C_K^1, then each equivalence class consists of Hecke characters of the form $\eta|\cdot|_{\mathbb{A}}^s$ with η a fixed unitary character, and a parametrization can be done to identify each class with complex plane, which shows that the set of Hecke characters has the structure of Riemann surface with countable connected components homeomorphic to $\mathbb{C}$, hence $\chi \mapsto \chi^\vee$ gives an biholomorphic involution on the space of Hecke characters.

2.4 Global Functional Equation

To formally define the integrand, a Haar measure on $\mathbb{A}^\times$ are fixed by

$$d^\times x = \prod_{v \nmid \infty} d^\times x_v \prod_{v|\infty} \frac{d^\times x_v}{1 - N(\mathfrak{p})^{-1}}$$

where $d^\times x = dx/|x|_v$ is the haar measure on $K_v^\times$. By part (b) of lemma 1.8, each $\mathcal{O}_v^\times$ gets measure $N(\mathfrak{d}_v)^{-1/2}$ with respect to measure $\frac{d^\times}{1-N(\mathfrak{p})^{-1}}$ for any finite place v, so for almost all finite place v, $\mathcal{O}_v^\times$ gets measure 1, so the deinition of Haar measure on idele group is well-defined by part (b) of lemma 2.1.

Definition 2.11. Let $f \in \mathcal{S}(\mathbb{A})$ and χ be a Hecke character, then the global zeta function associated to f and χ is defined by

$$Z(f, \chi) := \int_{\mathbb{A}^\times} f(x)\chi(x)\, d^\times x$$

We should restrict the domain to be one equivalence class of hecke characters, that means if we take a class $[\chi_0]$ with χ_0 a unitary Hecke character, then a parametrization should be

$$Z(f, \chi) = Z(f, \chi_0, s) = \int_{\mathbb{A}} f(x)\chi_0(x)|x|_{\mathbb{A}}^s\, dx$$

that allows us to view it as a integrand on complex plane.

Lemma 2.16. $\chi \mapsto Z(f, \chi)$ defines a holomorphic function on the domain of exponent $\sigma > 1$.

Proof. Write f as a product $\prod_v f_v$, then the inetgration ca be separted into product of local integrals, then there exists a finite subset S of places, which contains all archimedean palces, such that $f_v = \mathbf{1}_{\mathcal{O}_v}$ if $v \notin S$, then we have

$$|Z(f, \chi)| \leq \prod_{v \in S} |Z_v(f_v, |\cdot|_v^\sigma)| \cdot \prod_{v \notin S} |Z_v(f_v, |\cdot|_v^\sigma)|$$

where Z_v is the local zeta function at place v, the first term is finite product of local zeta functions, so by lemma 1.9 it will always nicely converge, so it is enough to estemate the second term by previous calcualtion

$$\begin{aligned}\prod_{v\notin S}|Z_v(f_v,|\cdot|_v^\sigma)| &= \prod_{v\notin S}\frac{1}{1-N(\mathfrak{p})^{-\sigma}}\\ &\leq \prod_{v \text{ finite}}\frac{1}{1-N(\mathfrak{p})^{-\sigma}} \quad (**)\\ &\leq \prod_{p \text{ prime}}(\frac{1}{1-p^{-\sigma}})^d = \zeta(\sigma)^d\end{aligned}$$

where $d=[K:\mathbb{Q}]$, and by analytical property of Riemann zeta function, the last term is finite for any $\sigma > 1$. □

Example 2.17 (Dedekind zeta function)**.** For any number field K, the Dedekind zeta function si defined by

$$\zeta_K(s) = \sum_{\mathfrak{a}\subset\mathcal{O}_K}\frac{1}{N(\mathfrak{a})^s}$$

where $\mathfrak{a}$ runs through all non-zero ideals of $\mathcal{O}_K$. It absolutely converges on half plane $\mathrm{Re}(s) > 1$ with Euler's product expansion

$$\zeta_K(s) = \prod_{\mathfrak{p}}\frac{1}{1-N(\mathfrak{p})^{-s}}$$

where $\mathfrak{p}$ runs through all prime ideals of $\mathcal{O}_K$, the proof is same with part $(**)$ in the proof.

A little work should be done to rewrite the global zeta function such that we can imitate the Riemann's proof of zeta function. Notice that $\mathbb{A}^\times/\mathbb{A}_1^\times \cong \mathbb{R}_{>0}$ as locally compact abelian groups, so by quotient integral formula (theorem 2.8) there exits a Haar measure d^*x on closed subgroup $\mathbb{A}_1^\times$ such that $d^\times x = d^*x \cdot dt/t$, so we can see the each $t > 0$ as an idele element by

$$\tilde{t} = (\underbrace{t^{1/N},\dots,t^{1/N}}_{\text{archimdean place}},1,...), \quad N=[K:\mathbb{Q}]$$

and then global zeta function can be rewritten as following

$$Z(f,\chi) = \int_0^\infty Z_t(f,\chi)\,\frac{dt}{t} \tag{1}$$

with auxiliary function defined by

$$Z_t(f,\chi) = \int_{\mathbb{A}_1^\times} f(\tilde{t}x)\,\chi(\tilde{t}x)d^*x$$

Hence now we can consider the symmerty of auxiliary function, it will be the key to finish the meomorphic continuation.

Lemma 2.18. $\chi \mapsto Z_t(f,\chi)$ defines a holomorphic function on the domain of Hecke characters, and for any $t > 0$ we have

$$Z_t(f,|\cdot|^s) = Z_{1/t}(\hat{f},|\cdot|^{1-s}) + \frac{\hat{f}(0)V}{t^{1-s}} - f(0)Vt^s$$

where V is the volume of fundamental domain of $\mathbb{A}_1^\times/K^\times$ with respect to the measure d^*x, and for any Hecke character χ not equivalent to $|\cdot|^s$, we have

$$Z_t(f,\chi) = Z_{1/t}(\hat{f},\chi^\vee)$$

Proof. Firstly we calculate the general equation of $Z_t(f,\chi)$, $K^\times$ can be identified as a discrete subgroup of $\mathbb{A}^\times$, so there exists a Haar measure $d^\star x$ on $\mathbb{A}_1^\times/K^\times$, such that

$$Z_t(f,\chi) = \int_{\mathbb{A}_1^\times/K^\times} \sum_{a\in K^\times} f(a\tilde{tx})\chi(a\tilde{tx})d^\star x$$

Hecke character is trival on $K^\times$, so $\chi(a\tilde{tx}) = \chi(\tilde{t}x)$ is independent of a, and we add the term $a=0$ such that we can apply the adelic Poisson summation formula

$$\begin{aligned} Z_t(f,\chi) + f(0)\int_{\mathbb{A}_1^\times/K^\times} \chi(\tilde{t}x)d^\star x &= \int_{\mathbb{A}_1^\times/K^\times} \sum_{a\in K} f(a\tilde{tx})\chi(\tilde{t}x)d^\star x \\ &= \int_{\mathbb{A}_1^\times/K^\times} \frac{1}{t|x|_{\mathbb{A}}} \sum_{a\in K} \hat{f}(a/\tilde{t}x)\chi(\tilde{t}x)d^\star x \end{aligned}$$

a change variable $x = y^{-1}$ allows us to calcluate the functional equation for any $t>0$

$$Z_t(f,\chi) + f(0)\underbrace{\int_{\mathbb{A}_1^\times/K^\times} \chi(\tilde{t}x)d^\star x}_{I(\chi)} = Z_{1/t}(\hat{f},\chi^\vee) + \hat{f}(0)\int_{\mathbb{A}_1^\times/K^\times} \chi^\vee(x/\tilde{t})d^\star x \tag{2}$$

hence it is enough to discuss the integration

$$I(\chi) = \chi(\tilde{t})\int_{\mathbb{A}_1^\times/K^\times} \chi(x)d^\star x$$

If $\chi = |\cdot|_{\mathbb{A}}^s$, then $I(\chi) = t^sV$ clearly; Otherwise, χ is not trival on $\mathbb{A}_1^\times/K^\times$ by lemma 2.14, and then compactness of $\mathbb{A}_1^\times/K^\times$ implies $I(\chi) = 0$. Finally, we plug the result into the equation 2 to finish the proof. □

Theorem 2.19 (Hecke-Tate-Iwasawa)**.**
Fix $f \in \mathcal{S}(\mathbb{A})$, then $\chi \mapsto Z(f,\chi)$ has a meomorphic continuation to the whole domain of Hecke characters, and it has only two simple poles at $|\cdot|^0$ and $|\cdot|^1$ with residue $-f(0)V$ and $\hat{f}(0)V$ respectively. Finally, it satisfies the functional equation

$$Z(f,\chi) = Z(\hat{f},\chi^\vee)$$

Proof. We separate the integral (equation 2.4) into two parts

$$Z(f,\chi) = \int_0^1 Z_t(f,\chi)\,\frac{dt}{t} + \int_1^\infty Z_t(f,\chi)\,\frac{dt}{t} \tag{3}$$

the second part is holomorphic on each component of character space since f decreases rapidly at infinity, so it is enough to evaluate the first part by changing variable $t \mapsto 1/t$, and applying the lemma 2.18 to arrange the integral. If $\chi = |\cdot|_{\mathbb{A}}^s$, then we can get

$$Z(f, |\cdot|_{\mathbb{A}}^s) = \frac{\hat{f}(0) \cdot V}{s-1} - \frac{f(0) \cdot V}{s} + \int_1^\infty [Z_t(f, \chi) + Z_t(\hat{f}, \chi^\vee)] \frac{dt}{t}$$

which gives the meomiorphic continuation of $\chi \mapsto Z(f, \chi)$ on the component of $|\cdot|_{\mathbb{A}}$, and the pole is given by first two terms; If χ is not of the form, similarly we can get

$$Z(f, \chi) = \int_1^\infty [Z_t(f, \chi) + Z_t(\hat{f}, \chi^\vee)] \frac{dt}{t}$$

which gives the holomorphic continuation on the other components. Finally, in both cases, the functional equation holds after a simple calculation. $\square$

Remark. A comparison can be done with the Riemann zeta function, the completed zeta function is defined by $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$, and a integral representation of ξ is given by

$$\xi(s) = \int_0^\infty \psi(t) t^{s/2} \frac{dt}{t}, \quad \mathrm{Re}(s) > 1$$

with

$$\psi(t) = \frac{1}{2}(\sum_{n \in \mathbb{Z}} e^{-\pi n^2 t} - 1)$$

and we can apply the poisson summation formula to get the symmetry of ψ, and then get the functional equation of ξ by diving the integral into two parts (see a proof in theorem 2.3 of chapter 6 in [5]). The Hecke-Tate theorem is a generalization of Riemann's proof, the role of auxiliary function $t \mapsto Z_t(f, \chi)$ is analogous to the role of ψ.

In complex analysis, the mellin transform of a function $f : \mathbb{R}_{>0} \to \mathbb{C}$ is defined by

$$\mathcal{M}[f](s) = \int_0^\infty f(t) t^s \frac{dt}{t}$$

the asymptotic behavior of f at 0 and ∞ determines a "analytical strip", where $s \mapsto \mathcal{M}[f](s)$ defines a holomorphic function, hence ...

If $\chi = \eta |\cdot|_{\mathbb{A}}^s$, then in equation (2.4) we notice that $\chi(\tilde{t}x) = \eta(x) t^s$, so a separtion of variable allows us to see the global zeta function on component $[\eta]$ as a mellin transform of a function on t as following

$$\Phi(t) = \int_{\mathbb{A}_1^\times} f(\tilde{t}x) \eta(x) \, d^*x$$

3 Conclusion

A overview of the backgroud and the history related to Tate's thesis is given in this section to make a conclusion.

The conceptual trajectory of this functional equation can be concluded as Riemann - Hecke - Tate. Riemann (1859, [3]) gave the first proof of the functional equation of Riemann zeta function, after that Dedekind generalizes the Riemann zeta function to number fields, and Hecke [6] gives a proof of the functional equation of Dedekind zeta function. Hecke consider the theta function associated to an ideal class, and use the poisson summation formula on $\mathbb{R}^n$ to finish the proof, a proof of such style can be found in chapter 13 of [7].

In number theory, the discussion about the L-functions is unavoidable, and before Hecke's proof the functional equation of dirichlet L-function has been understood well as following

Theorem 3.1. For any primitive Dirichlet character χ modulo $q > 1$, the completed L function $\Lambda(s,\chi) = (q/\pi)^{\frac{s+\mathfrak{v}}{2}}\Gamma(\frac{s+\mathfrak{v}}{2})L(s,\chi)$ has an analytic continuation to the whole complex plane, and it satisfies the functional equation

$$\Lambda(1-s,\bar{\chi}) = W(\chi)\Lambda(s,\chi)$$

where $\mathfrak{v} = \frac{1-\chi(-1)}{2}$ and $W(\chi)$ is the root number

$$W(\chi) = \frac{1}{i^{\mathfrak{v}}\sqrt{q}} \underbrace{\sum_{a=1}^{q} \chi(a) e^{2\pi i a/q}}_{\text{Gauss sum}}$$

For general L-functions on number field, there are at least two problems to consider the functional equation, the first one is what characters is proper to choose, and the other one is how to use poisson summation formula. Hecke (1918, [8]) intorduced a type of characters as following

Definition 3.1. Let $\mathfrak{m}$ be an ideal of $\mathcal{O}_k$, a **Größencharakter modulo** $\mathfrak{m}$ is a character $\chi : J(\mathfrak{m}) \to \mathbb{S}^1$ for which there eixsts a pair of (continous) characters

$$\chi_{\mathfrak{m}} : (\mathcal{O}_k/\mathfrak{m})^* \to \mathbb{S}^1, \quad \chi_\infty : k_\infty^* \to \mathbb{S}^1$$

such that

$$\chi((a)) = \chi_{\mathfrak{m}}(\bar{a}) \cdot \chi_\infty(i(a))$$

for any $a \in k(\mathfrak{m}) \cap \mathcal{O}_k$, where $J(\mathfrak{m})$ is all fractional ideals of $\mathcal{O}_k$ coprime to $\mathfrak{m}$, $k(\mathfrak{m})$ is the set of all elements in k coprime to $\mathfrak{m}$, and $i : k \to k_\infty$ is the minkowski embedding.

which allows him to define the Hecke L-function associated to a Größencharakter χ by

$$L(s,\chi) = \sum_{\mathfrak{a}} \chi(\mathfrak{a}) N(\mathfrak{a})^{-s}, \quad \mathrm{Re}(s) > 1$$

and then finished the proof of functional equation by a complicated theta function, a proof of such style can be found in the chapter 12 of [7].

The Hecke's proof is complicated, the reason is that the strcuture of Größencharakter will depend on the unit group $\mathcal{O}_K^\times$. After Chevalley (1938) pushforward the concept of idele group, new language of number theory was developed to cover the clssical algebraic number thoery, and in this framework Tate (1950, [1]) gives a new proof of the functional equation of Hecke L-functions. It should be noticed that Iwasawa (1950, [9])independently developed a similar adelic proof at the same time, and his work deserves being mentioned here.

It is meaningful to think about why Hecke characters are the generalization of Größencharakter, actually they are all the generalization of dirichlet characters, a clear discussion can be found in [10].

This section will end by the proof of Hecke's L-function, which answers another natural question: what is the relation between local and global zeta functions?

Definition 3.2. Let χ be a unitary Hecke character, and S is the finite subset containing all archimedean places such that χ_v is unramified for any $v \notin S$, then the partial L-function associated to χ is defined by

$$L_S(s,\chi) := \prod_{v \notin S} \frac{1}{1-\chi(\mathfrak{p}_v)N(\mathfrak{p}_v)^{-s}}$$

with convention that $\chi(\mathfrak{p}_v) = \chi_v(\varpi_v)$.

If χ is trivial, tnen S is just the set of all archimedean places, and $L_S(s,\chi)$ is exactly the Dedekind zeta function of K, and for other case

$$\sum_{v \in S} |\chi(\mathfrak{p}_v)N(\mathfrak{p}_v)^{-s}| = \sum_{v \in S} |N(\mathfrak{p}_v)|^{-\mathrm{Re(s)}} \leq \zeta_K(\mathrm{Re}(s))$$

which allows u to conclude that L-partial L-function is holomorphic on half plane $\mathrm{Re}(s) > 1$. Applying lemma 1.14, we can write partial L-function as a product of local zeta functions as following

$$\prod_{v \notin S} N(\mathfrak{d}_v)^{-1/2} \cdot L_S(s,\chi) = \prod_{v \in S} Z_v(\mathbf{1}_{\mathcal{O}_v}, \chi_v|\cdot|^s)$$

The first term is finite since almost local extension is unramified, so we can enlarge the domain of S to conclue more non-trivial information.

Definition 3.3. A finite set S_∞ is defined as following: it contains all archimedean places, and for any $v \notin S_\infty$, (1) χ_v is unramified; (2) the local extension $K_v/\mathbb{Q}_p$ is unramified; (3) the conductor of standard character ψ_v is $\mathcal{O}_v$. The L-function associated to χ is defined by

$$L(s,\chi) = \prod_{v \notin S_\infty} \frac{1}{1-\chi(\mathfrak{p}_v)N(\mathfrak{p}_v)^{-s}}$$

Theorem 3.2 (Hecke-Tate-Iwasawa)**.**
For a unitary Hecke character χ, the L-function $L(s,\chi)$ has an analytic continuation to the whole complex plane, and it satisfies the functional equation

$$L(s,\chi) = \prod_{v \in S_\infty} \rho_v(\chi_v|\cdot|_v^s) \cdot L(1-s,\chi^{-1})$$

where ρ_v is the local ρ-factor defined in the section 1.3.

Proof. By above argument, we can choose some proper functions $f_v \in \mathcal{S}(K_v)$ for any $v \in S$, and $f_v = \mathbf{1}_{\mathcal{O}_v}$ if $v \notin S$, such a function is defined as f and it is a schwartz-bruhat function on $\mathbb{A}$ by definition. Then we consdier the global zeta function $s \mapsto Z(f,\chi|\cdot|_{\mathbb{A}}^s)$ on the component $[\chi]$, if $\mathrm{Re}(s) > 1$ then global zeta function can be written as a product of local zeta functions

$$Z(f,\chi|\cdot|_{\mathbb{A}}^s) = \prod_v Z_v(f_v,\chi_v|\cdot|_v^s) = \prod_{v \in S} Z_v(f_v,\chi_v|\cdot|_v^s) \cdot L(s,\chi)$$

which allows us to write

$$L(s, \chi) = \frac{Z(f, \chi|\cdot|_{\mathbb{A}}^{s})}{\prod_{v \in S} Z_v(f_v, \chi_v|\cdot|_v^{s})}$$

the numerator has a meomorphic continuation to the whole complex plane by theorem 2.19, and the denominator is a finite product of local zeta functions, it has a meomorphic continuation to a riemann surface by theorem 1.15. Hence $s \mapsto L(s, \chi)$ has a meomorphic continuation, and apply the functional equation of global and local zeta functions, we can finish the proof. $\square$

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